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Nanoscale Structure and Interaction of Condensed Phases of DNA–Carbon Nanotube Hybrids

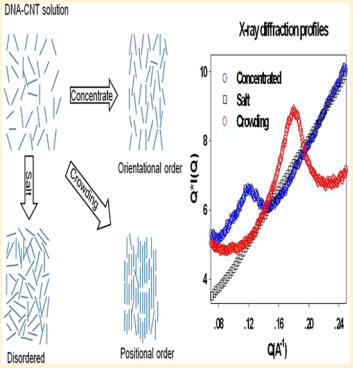
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Supporting Information

ABSTRACT: Condensation of DNA–carbon nanotube (CNT) hybrids dispersed in aqueous solutions can be induced by elevated hybrid concentrations, salts, or crowding agents. DNA–CNT condensates exhibit either nematic ordering or amorphous aggregates, dependent on the nature of interhybrid interactions. This study employed X-ray diffraction (XRD) to determine nanoscale structures of the condensates, including the presence of positional ordering, interaxial distances, and the range of ordered domains. To probe the effects of DNA sequence, two types of CNT hybrids, dispersed by genomic DNA of random sequence and synthetic oligonucleotides respectively, were studied under identical conditions. The osmotic stress method was further used to quantify force–distance dependencies of the DNA–CNT hybrids to elucidate the relation between interhybrid interactions and condensate structures. We observed that, independent of DNA sequence, lyotropic DNA–CNT phases showed weak positional ordering with long interhybrid distances, salt-induced condensates were amorphous, crowding-condensed DNA–CNTs were the most ordered with pronounced XRD peaks, and interhybrid interactions were defined by short-range hydration repulsion and long-range electrostatic repulsion. Conversely, the effects of DNA sequence became evident as to their quantitative force–distance relationships. Genomic DNA of random sequence consistently gave longer interhybrid distances than synthetic oligonucleotides, which we attribute to the likely differences in their hybrid diameters.



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1. INTRODUCTION

Single walled carbon nanotube (CNT) has attracted much attention owing to its outstanding mechanical, thermal, optical and electric properties. However, because of its strong van der Waals interaction, CNT, despite being a nanometer-sized rod-like molecule, exists in aggregate forms difficult to process. To overcome this barrier, liquid-phase dispersions of CNTs have been explored employing diverse strategies ranging from strong acids[1] and nonaqueous solvents[2−4] to chemical modification[5] to noncovalent absorption with amphipathic molecules.[6−10] Among them, the method of noncovalent absorption is arguably the most studied as it is easy to implement and can retain the original structure and properties of CNTs. As a result, liquid-phase processing of CNTs has promoted broad applications in fiber spinning,[11] field-effect transistor,[12] chemical sensor[13] and biomedicine.[14] Among the noncovalent dispersion agents, DNA emerged as one of the most effective[15] as to dispersion capacity and stability, as well as the potential for biological applications. It is generally accepted that DNA wraps around the CNT via $\pi-\pi$ stacking interaction between the nucleobases and CNT[16,17] and the exposed DNA phosphate backbones bestow polyelectrolyte-like properties to the DNA–CNT hybrids. As such, the phase behaviors and processing characteristics of DNA–CNT hybrids have been under active investigation.

The predominant feature of DNA–CNT phase behaviors is perhaps the formation of liquid crystal (LC) phase due to its large length/diameter ratio and rigidity, e.g., its persistence length is larger than 32 μm.[18] The LC phase is considered a key precursor of the fluid-phase processing into fibers or composite materials with superior properties,[19] supporting the significance of its study. It is worth noting that LC phases have also been reported for CNTs dispersed in other systems such as acids,[20] thermal responsive gel,[21] and bile salts.[22] For DNA–CNT hybrids, a series of pioneering studies from the groups of Poulin, Davis and others have mapped out DNA–CNT phase diagrams in aqueous solutions as functions of DNA–CNT concentration, salt concentration, and the addition of neutral polymers.[23−25] Assuredly, DNA–CNT hybrids exhibit phase behaviors as expected from highly anisotropic and charged rod-like molecules. Upon increasing the hybrid concentration, spontaneous phase separation gives rise to lyotropic nematic LC phases. In the presence of molecular crowding agents (e.g., neutral polymers), nematic LC phases can also develop. Note that this crowding-induced condensation depends on the CNT length arising from length-dependent depletion forces, which has led to a convenient method of CNT length fractionation.[26]

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Salt acts in a qualitatively different way as it mainly works to screen the electrostatic repulsion between like-charged DNA−CNT hybrids. Salt thus typically facilitates the formation of LC phases, except at high concentrations (to be discussed later).

Besides the anisotropic geometry, repulsive interaction between DNA−CNT hybrids is considered critical for the formation of LC phases. For example, monovalent salts at >1 M concentration are able to condense DNA−CNTs, reminiscent of the "salting-out" process of proteins, but such condensates do not show LC ordering and instead exist as quasi-random percolated networks of "sticky" rods.[24] The lack of ordering was attributed to the net attraction between DNA−CNT hybrids under high salt.[27,28] Nonetheless, this is somewhat in contrast with the behavior of another well-studied rod-like polyelectrolyte, double-stranded DNA (dsDNA), where multivalent counterion induced dsDNA condensates exhibit high degrees of LC ordering.[29] Answers to their disparate behaviors may lie in the physical origin of interhybrid attraction which likely differs from that of inter-dsDNA attraction. In a recent effort, Tardani et al. explored the phase behaviors of DNA−CNTs in the presence of polycationic molecules and reported interesting commonalities and differences of DNA−CNTs compared with other polyelectrolytes.[30] Mechanistic studies are called for to fully elucidate the relation between interhybrid interactions and resultant condensate structures.

In this study, we applied X-ray diffraction (XRD) as the primary experimental technique to analyze nanoscale structures of the condensed phases of DNA−CNT hybrids. Their nanoscale structures are largely unknown as the two most commonly used techniques in previous studies, polarization light microscopy (PLM) and atomic force microscopy (AFM), provide information on either microscopic alignment or local topology, whereas XRD probes the ensemble structures of DNA−CNT condensates at the nanometer scale. Note that there exist a large body of XRD studies of CNT fibers and composites synthesized with various methods and our discussion here is limited to the cases of spontaneous condensation of CNTs dispersed in aqueous solutions. To our knowledge, the hexagonal packing of DNA−CNT LC phases was first reported by our group via XRD measurements of capillary-aligned and crowding-condensed DNA−CNTs.[31] Given XRD's unique utility to provide structural information such as the degree of nanoscale ordering and interhybrid spacing, this study extends XRD measurements to the condensed phases of DNA−CNTs as functions of hybrid concentration and salt valence and concentration.

The other main component of this study is to examine the potential roles of DNA sequence in modulating interhybrid interactions and subsequently hybrid phase behaviors. As an initial step, we specifically compared CNTs dispersed by denatured genomic dsDNA (as random sequence dsDNA−CNT hybrids) and oligomeric synthetic single-stranded DNA (as uniform sequence ssDNA−CNT hybrids). DNA sequence matters as to DNA binding affinity and DNA conformation on CNT surfaces, which can lead to different charge densities and surface biochemistries. Such dependences are also regarded to underlie DNA-assisted CNT sorting methods based on peculiar recognitions between short ssDNAs of specific sequences and CNTs of specific chiralities.[32,33] Elucidating the roles of DNA sequence in modulating hybrid behaviors is therefore of both scientific and practical significance. Here the denatured genomic dsDNA and synthetic short ssDNA represent two extreme classes of DNA sequences and are instructive as model systems. For this purpose, in addition to XRD analyses of the condensed phases of their respective CNT hybrids as described above, we further applied the osmotic stress method to measure the interhybrid force as a function of interhybrid distance to quantitate interhybrid interactions. Note that we have previously determined the force−distance relationship of the same ssDNA−CNT hybrids at low salts[31] but not the dsDNA−CNT hybrids.

Specifically, this study used denatured calf thymus dsDNA and synthetic $(GT)_{20}$ ssDNA to disperse CNTs predominantly of (6,5) chirality. PLM was used to check the LC ordering of the condensed phase and the phase behaviors were found to be consistent with the literature. XRD was used to analyze their condensed phases across broad ranges of hybrid concentrations (1−50 mg/mL), salt type and concentration (NaCl up to 2 M, $MgCl_2$ up to 100 mM, and $SpermineCl_4$ up to 10 mM), and added neutral polymer (poly(ethylene glycol) 8000 Da or PEG, 0 to 50% by weight). We found that, at hybrid concentrations below 30 mg/mL, though PLM indicated LC ordering, no XRD peaks could be identified. Salt-induced hybrid condensation, regardless of ion valence, lead to amorphous condensates without XRD features. PEG-induced condensates showed the highest degrees of LC ordering and sharpest XRD peaks. The force−distance dependences of dsDNA−CNT and ssDNA−CNT showed remarkable similarities in curve shapes, as well as notable differences in interhybrid distances. We discuss our observations in terms of hybrid polydispersity, the nature of hybrid interactions, and the differences between dsDNA and ssDNA.

2. MATERIALS AND METHODS

CNT (CoMoCAT SG65i) was obtained from SouthWest NanoTechnologies and chosen for its high enrichment of (6,5) chirality CNTs (>40%). Single-stranded $(GT)_{20}$ DNA was obtained from Integrated DNA Technologies and chosen as a model ssDNA for its strong binding to CNT.[34] Calf thymus dsDNA, poly(ethylene glycol) (PEG, M = 8000 g/mol) and sodium chloride were purchased from Sigma and used as received. Preparation of DNA−CNT hybrids followed the established protocols[35] and is briefly described below. First, DNA stocks were dissolved in 1× TE buffer (10 mM Tris, 1 mM EDTA, pH 7.5) and CNTs in powder form were sonicated in deionized water for 10 min with a Sonics VCX130 unit (3 mm-diameter probe, power of 5 w). The dsDNA stock was further denatured by heating to 95 °C followed by rapid cooling. DNA and CNT mixtures (DNA−CNT of ∼1 mg/mL, the weight ratio of DNA/CNT is 2 for both dsDNA and ssDNA) were then sonicated in ice bath for 4 h at a power of 8 w, followed by centrifugation of 14 000 rpm for 3 h to remove impurities and undispersed CNTs. For convenience, the hybrids are referred as dsDNA−CNT and ssDNA−CNT for CNTs dispersed by calf thymus dsDNA and $(GT)_{20}$, respectively. To determine the DNA−CNT concentrations, due to the dependences of CNT optical absorption on many factors such as length and chiral distribution,[36] the concentrations of free DNA (i.e., unbound to CNTs) were first measured by optical absorption at 260 nm after condensing DNA−CNTs by salt and centrifugation, the concentrations of bound DNA were then obtained given the known total DNA concentrations. Taking the weight ratio of bound-DNA:CNT was taken as 1.2:1,[36] these measurements gave concentrations of ∼0.47 and ∼0.55 mg/mL for the dsDNA−CNT and ssDNA−CNT stock solutions used, respectively. These values

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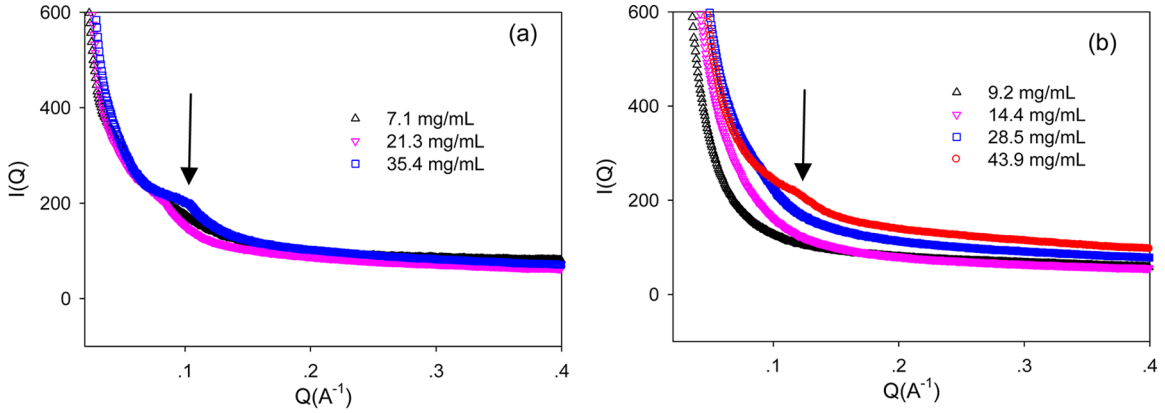


Figure 1. XRD curves of the hybrid condensates at varied nominal DNA−CNT concentrations: (a) dsDNA−CNT; (b) ssDNA−CNT. The insets show the solution concentrations, and the arrows indicate the relatively weak XRD peaks.

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were found to give the conversion of 17 μg/mL CNT for an optical density of one at 990 nm (the E_{11} peak of (6,5) CNT), which was used to determine the DNA−CNT concentration in subsequent experiments via optical absorptions.

PLM measurements were performed on an AmScope XY-P unit using precleaned glass slides and coverslips. The images were taken in a transmission mode and the magnification was varied between 4, 10, 40, and 60 times. XRD measurements were carried out at 20 °C with an in-house instrument which integrates a microfocus fixed Cu-anode source and an image plate detector. Data analysis was done with home-written Matlab programs.

3. RESULTS AND DISCUSSION

As the main structural technique of this study, XRD is sensitive to the presence of structural ordering in nanometer scales. In the case of DNA−CNT LC phases, the parallel hybrid alignment leads to a primary XRD peak informing the interaxial distance d. Taking into account of the hexagonal geometry, d is given by $2\pi/Q_{peak} \times 2/\sqrt{3}$, where Q is the scattering factor defined as $4\pi\sin(\theta)/\lambda$ with θ as half of the scattering angle and λ as the X-ray wavelength. Note that the DNA−CNT condensates studied here do not have sufficient ordering to show prominent higher order XRD peaks previously observed by us in capillary-aligned DNA−CNTs,[31] and that the presumption of hexagonal ordering should be considered as empirical. In addition to the peak position, the peak width σ gives the apparent size of ordered domains via the Scherrer formula $D = 2\pi/\sigma$, which can also be understood as the correlation length of DNA−CNT alignment.

We first investigated the lyotropic behaviors of DNA−CNT hybrids by concentrating as-prepared dispersions via centrifugal filtration. Filtration is advantageous in maintaining constant salt condition (1 × TE buffer) compared with evaporative methods, and the sample concentration can be conveniently determined by monitoring the weight changes of the filtered solution. For both dsDNA−CNT and ssDNA−CNT hybrids, hybrid condensates appeared at ∼4 mg/mL (total weight of starting hybrids divided by total solution weight, based on a constant density of 1 mg/mL). XRD was taken starting at 7 mg/mL (i.e., when sufficient amount of condensates was available) up to ∼50 mg/mL above which the centrifugal method became ineffective. PLM was also used to verify the LC nature of the condensates (Supporting Information, Figure S1). XRD profiles of DNA−CNT condensates at selected concentrations are shown in Figure 1. It is somehow puzzling that none or very weak XRD peaks were observed, despite clear LC ordering by PLM. The only exception was the dsDNA−CNT condensate at the highest concentration (51.5 mg/mL) that showed a rather strong XRD peak, the molecular origin of which was however unclear (see Supporting Information, Figure S2 for data and discussions). Overall, we attribute this lack of XRD features to the relatively long interaxial distances in these condensates. Taking dsDNA−CNT at 35.4 mg/mL for an example (Figure 1a and Supporting Information, Figure S3a), the Q_{peak} of 0.1 Å^{-1} yields an interaxial distance of 72.5 Å and surface−surface separation over 42.5 Å (taking an upper limit of 30 Å for the hybrid diameter). It is thus not surprising that the spaced-out dsDNA−CNTs would quickly lose positional ordering while still maintaining excellent orientational ordering due to their long lengths. The same arguments also apply to the case of ssDNA−CNTs showing broad XRD peaks at comparable positions (Figure 1b and Supporting Information, Figure S3b).

Next, salt-induced DNA−CNT condensation was investigated by adding concentrated salt solutions to the CNT dispersions. Cations are able to effectively reduce the repulsion between negatively charged DNA−CNT hybrids, and multivalent ions may even mediate interhybrid attraction by analogy with inter-dsDNA attraction mediated by multivalent cations (valence ≥3).[37] We thus varied both the salt concentration (up to 2 M) and cation-valence (1+ to 4+). Monovalent salt (NaCl) was observed to condense both dsDNA−CNT and ssDNA−CNT above 1 M, while much lower concentrations were observed for divalent ($MgCl_2$ > 10 mM) and tetravalent (Spermine^{4+} > 0.6 mM) ions (see Supporting Information, Figure S4). Here we present the results of the ssDNA−CNT only due to the possible complications of the dsDNA−CNT as discussed in Supporting Information, Figure S2, e.g., multivalent-cation condensed dsDNA−CNTs showed strong XRD peaks reminiscent of condensed dsDNA. None of the salt-induced ssDNA−CNT condensates exhibited significant birefringence or XRD peaks (Supporting Information, Figure S5), indicative of amorphous aggregates of rod-like hybrids. The aggregates likely arise from strong interhybrid adhesion that arrests the condensate in (meta)stable disordered arrangements, and the XRD study here provides valuable confirmation of the lack of nanoscale ordering in DNA−CNT condensates. The fractal dimensions of the formed aggregates were significantly larger than 1 when $MgCl_2$ and spermine were

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added (Supporting Information, Figure S 5d), indicating that multivalent ions promote lateral growth of aggregates.

The third series of experiments examined molecular crowding induced DNA−CNT condensates. We have previously studied crowding-induced ssDNA−CNT condensates with XRD and showed that they exhibit high degrees of LC ordering.[31] This study extended to dsDNA−CNT condensates and, more importantly, aimed to illustrate DNA-sequence dependent structures and interactions of the hybrid condensates. With XRD to elucidate nanoscale structures, interhybrid interactions were quantified in terms of force−distance relationships via the osmotic stress method (OSM). The OSM is based on the same physical principle as molecular crowding in this context[38] where neutral osmolytes exert depletion forces to push DNA−CNTs together into macroscopic LC phases. At equilibrium, neighboring DNA−CNTs are pushed until the distance at which the force between DNA−CNTs is the same as the osmolyte-induced depletion force. By varying the osmolyte concentration and subsequently the depletion force, concurrent XRD determination of the interhybrid distance yields the force−distance relationship. For the OSM experiments, 120 μL DNA−CNT solution was added to 120 μL PEG 15% (w/w) NaCl 300 mM solution, vortexed and centrifuged at 20 000 × g for 30 min to obtain one pellet. Note that ssDNA or dsDNA does not spontaneously condense under this condition, and thus, the obtained pellet was thus free of pure DNA phases. Then the pellets were taken out of supernatant and equilibrated in PEG solutions of different concentrations at varied NaCl concentrations (150, 300, and 500 mM). The pellets can be redissolved in low salt buffers and then condensed again by PEG to give the same X-ray diffraction peak features. Our previous studies have shown that an incubation time of 2 weeks were sufficient to reach phase equilibrium.[31] Thus, one solution exchange was performed over a period of 2 weeks here before PLM and XRD measurements. LC ordering was observed in all samples via PLM (Supporting Information, Figure S6), while the LC domains appeared more granular compared with lyotropic DNA−CNT condensates.

As shown in Figure 2 and Supporting Information, Figure S7, both PEG-condensed dsDNA−CNTs and ssDNA−CNTs exhibit well-defined diffraction peaks indicative of excellent LC ordering regardless of DNA sequence. However, details of the peak profiles differ. Under 15% (w/w) PEG solution in 300 mM NaCl, the XRD position is at 0.168 Å^{-1} for dsDNA−CNT and 0.181 Å^{-1} for ssDNA−CNT, giving the interaxial spacing of d=43.2 and 40.1 Å, respectively. This reveals a difference of 3.1 Å with dsDNA−CNT being farther apart under the same crowding and salt conditions. The intermolecular distance is expected to be reduced by increasing external osmotic pressure. Accordingly, under 35% PEG solutions, the peak positions shifted to higher Qs, 0.201 Å^{-1} for dsDNA−CNT and 0.223 Å^{-1} for ssDNA−CNT, resulting in d = 36.0 and 32.5 Å, respectively. This demonstrated that osmotic pressure could be a very effective tool to control the nanoscale structures of the LC phase with high precision, which may be closely related to the properties of CNT-based composite materials. In addition, the widths of XRD peaks provide information on the spatial range of structural ordering. For both dsDNA−CNT and ssDNA−CNT condensates, the peaks in Figure 2 and those under other PEG solutions between 10% and 50% gave a comparable full width at half-maximum σ of 0.050 Å^{-1} (Supporting Information, Figure S8) which corresponds to ∼126 Å Scherrer domain size. This indicated that the crowding-induced condensates lost ordering just over a few interaxial distances, likely explaining the absence of higher order XRD peaks. On the whole, XRD structural analyses indicate substantial commonalities between dsDNA−CNT and ssDNA condensates, as well as a notable difference of larger interaxial distances for dsDNA−CNTs.

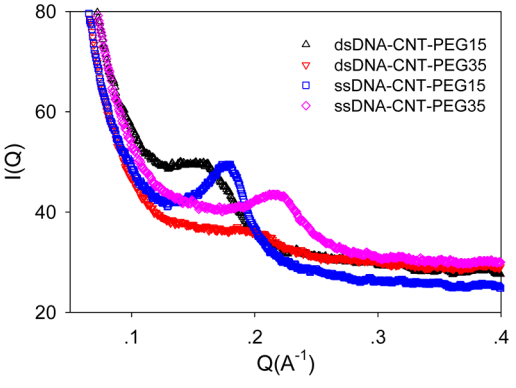


Figure 2. XRD curves of DNA−CNT in PEG solutions at 300 mM NaCl. The inset shows the sample information, e.g. "dsDNA−CNT-PEG15" means dsDNA−CNT samples equilibrium in 15% PEG solution.

Given that the difference in the interaxial distances necessarily results from the difference in DNA wrapping, the force−distance dependencies may provide further insight. Figure 3 shows such curves for dsDNA−CNT and ssDNA−CNT at varied salt concentrations. The differences in interaxial distances appeared to persist regardless of PEG or salt concentrations. The curves from dsDNA−CNT and ssDNA−CNT, however, showed very similar shapes with the main difference being an offset in interaxial distances (Figure 3a−c). Indeed, as shown in Figure 3d, shifting dsDNA−CNT curves by −3.3 Å results in excellent agreements with that of ssDNA−CNT. Their close resemblances in force−distance dependencies suggested common physical nature of the interhybrid interactions. The salt dependence of the force curves (Figure 3d) indicated two distinct regimes, a largely salt-independent short-range repulsion and a salt-screened longer-range repulsion. This was consistent with our previous studies of ssDNA−CNT forces that were shown to be dominated by repulsive hydration force at short-range and electrostatic repulsion at medium and long ranges,[31] and this study showed that dsDNA−CNT interactions had the same physical origins.

One difference yet to be explained is the larger interaxial distance between dsDNA−CNTs than that between ssDNA−CNTs under the same conditions. In light of their common physical nature of interhybrid interactions, one possible explanation is the difference in charge density, specifically the larger charge density of dsDNA−CNT than ssDNA−CNT. However, as the $(GT)_{20}$ ssDNA sequence studied here is among the most effective CNT dispersant, it was unlikely that denatured genomic dsDNA could match its CNT binding capacity. Another possible explanation is that the dsDNA−CNT hybrid had a larger diameter than the ssDNA−CNT hybrid. The larger diameter could come from the looser wrapping of the random sequence of denatured dsDNA than the specific sequence $(GT)_{20}$ ssDNA, and the longer length of the denatured dsDNA (∼300 bases after sonication[25]) may also lead to looser wrapping as reported by Yang et al. in their study

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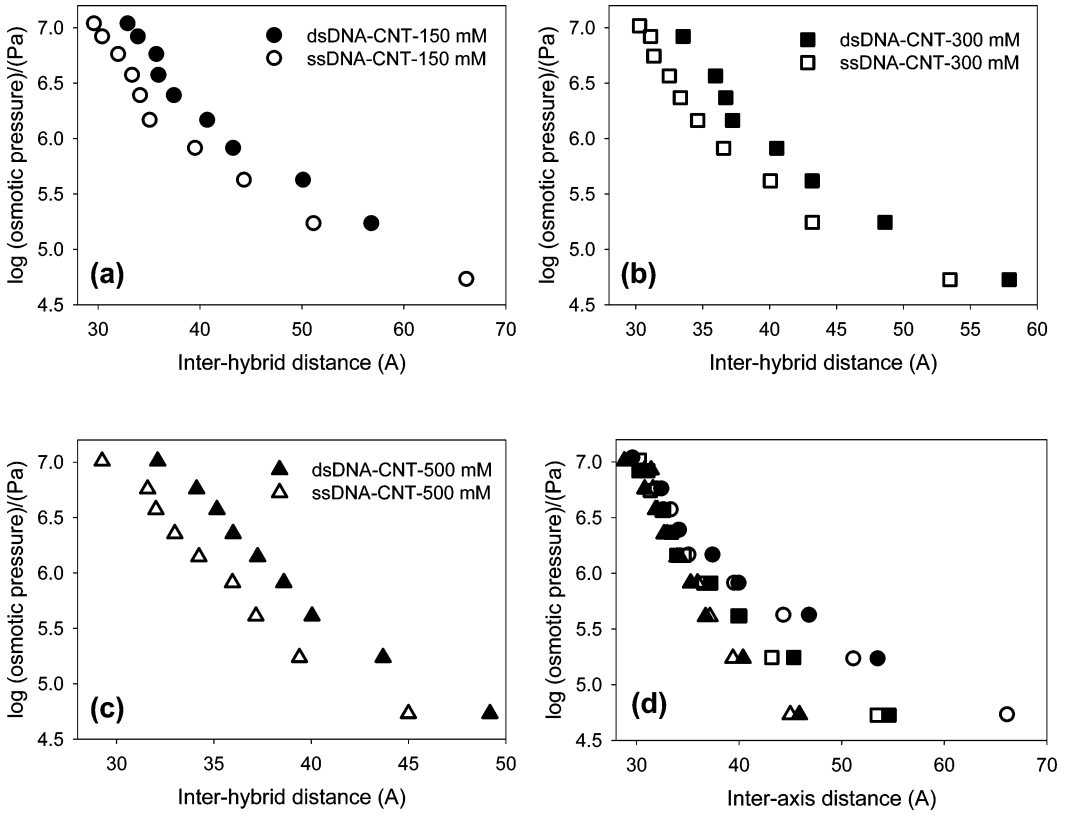


Figure 3. Comparison of force−distance curves between dsDNA−CNT and ssDNA−CNT at various NaCl concentrations: (a) 150 mM; (b) 300 mM; (c) 500 mM. (d) Force distance curves of DNA−CNT in PEG solutions at various NaCl concentrations. The data of dsDNA−CNT were shifted to the left by 3.3 Å along the *x* axis. The legends to the symbols are the same as that in parts a−c.

of DNA length dependence.[39] The presence of incompletely denatured dsDNA on the CNT surface, though expected to be negligible in our study, offers another possible cause for a larger diameter for dsDNA−CNT.[17] While further studies are necessary to elucidate the origin of the differences, XRD in conjunction with the OSM shows to be sensitive to subnanometer scale variations of DNA−CNT structures and may serve as a quality control method for CNT dispersions.

4. CONCLUSIONS

Employing XRD and OSM to probe nanoscale structures and intermolecular interactions, we have analyzed DNA−CNT condensates obtained by increasing DNA−CNT concentration, adding salts, and adding neutral crowding polymers. Two distinctive DNA−CNT hybrids, prepared with denatured genomic dsDNA and synthetic oligonucleotides respectively, were contrasted to probe the effect of DNA sequence on condensate structures and interactions. While the phase diagrams of DNA−CNTs have been largely mapped out, XRD provides unique information on the presence of positional ordering at the nanometer scale, interaxial distances, and the range of ordered domains. Lyotropic DNA−CNT condensates exhibited high degree of orientational ordering but weak positional ordering due to their long interaxial distances. Salt-induced DNA−CNT condensates do not show positional ordering in consistency with previously reported amorphous aggregates. Polymer-condensed DNA−CNTs gave rise to LC phases of best positional ordering, and enabled the determination of force−distance relationships as formulated by the OSM. In line with the similar nanoscale structures of dsDNA−CNT and ssDNA−CNT condensates, their interhybrid interactions also resemble each other with defining features of short-range hydration repulsion and medium/long-range electrostatic repulsion. Despite all the commonalities, dsDNA−CNT and ssDNA−CNT consistently show different interaxial distances under the same crowding and salt conditions. A likely cause is a larger hybrid diameter of dsDNA−CNT due to its random DNA sequence, as well as its longer DNA length. In light of the potential applications of CNTs in condensed forms, this study shows that solution conditions can effectively modulate the nanoscale structures of such condensates of importance to their functional properties.

We find the physical origin of monovalent-salt-induced DNA−CNT condensation particularly intriguing. While the "salting-out" of proteins is well established, monovalent salts cannot condense double-stranded DNA which is expected to mimic more closely the DNA−CNT surface than proteins. Knowledge of the exact conformation of the wrapped DNA would provide valuable insight into the origin of salt-induced (effective) interhybrid attraction. The ability to measure the interhybrid interactions under condensing conditions would also be very useful. From a fundamental perspective, DNA−CNT hybrids may serve as a unique model system as a rigid rod-like polyelectrolyte with tunable surface chemistry (i.e., DNA sequence and CNT chirality) for future studies.

■ ASSOCIATED CONTENT

Supporting Information

PLM images and some XRD profiles of DNA−CNT, $Q^*I(Q) \sim Q$ plots, and differential sedimentation assay. The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.5b04794.

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Notes

The authors declare no competing financial interest.

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Supplemental Data

Nanoscale Structure and Interaction of Condensed Phases of DNA-Carbon Nanotube Hybrids

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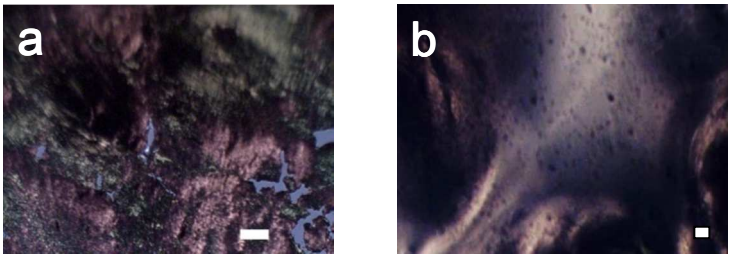


Figure S1. Polarized light microscopy (PLM) images of DNA-CNT condensed phases that form spontaneously upon increasing hybrid concentrations. The scale bars are 100 μm. (a) 21.3 mg/mL dsDNA-CNT; (b) 28.5 mg/mL ssDNA-CNT.

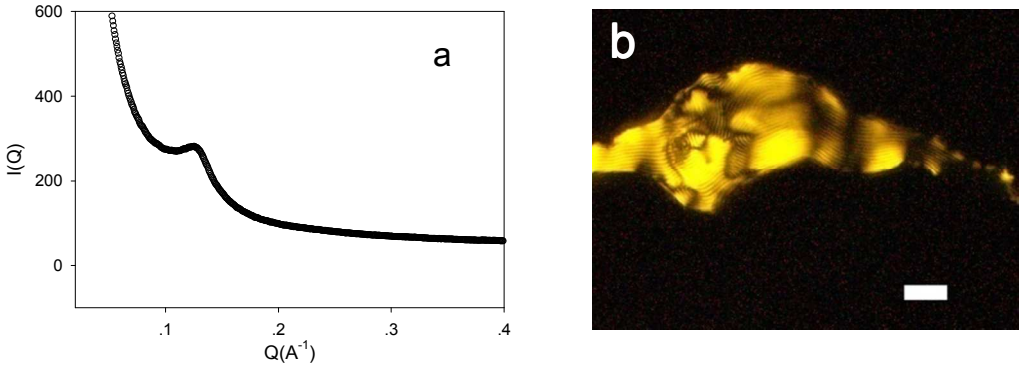


Figure S2. XRD and PLM data of the dsDNA-CNT condensate at 51.5 mg/ml nominal hybrid concentration. (a) The XRD profile indicating a pronounced peak around 0.12 $Å^{-1}$. (b) The PLM image demonstrating a cholesteric LC phase. The scale bar is 20 μm. The molecular origin of the XRD and PLM features is complicated by the presence of excess amount of DNA in as-dispersed dsDNA-CNT solutions (dsDNA:CNT=2:1 by mass). Even after sonication, the free DNA fragments, between 200 and 750 bases[1], do not pass through the filter, e.g., 200 bases of ssDNA has molecular mass of ~60 kilo Dalton. While the fragments are denatured, they have random sequences and it is plausible that some fragments or part of them hybridize into double-stranded forms that can form cholesteric LC phases at high concentration[2]. This makes it difficult to determine the molecular identity of observed features.

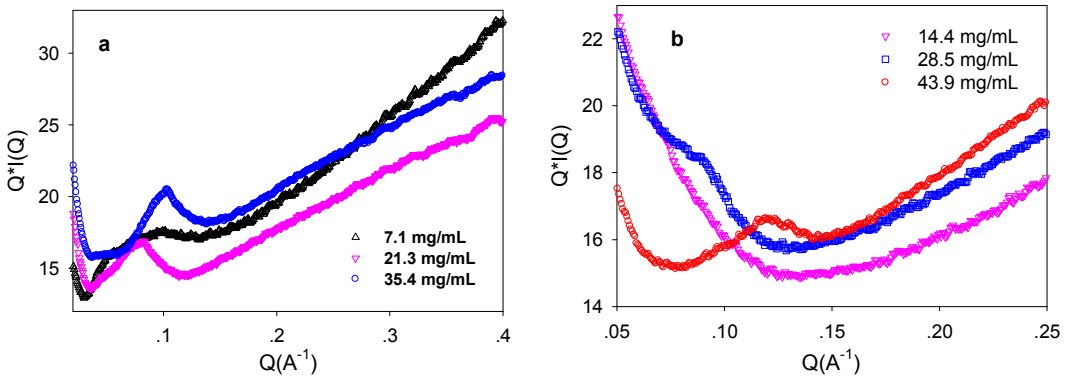


Figure S3. Q*I(Q)~Q plots of the hybrid condensates at varied nominal DNA-CNT concentrations. (a) dsDNA-CNT; (b) ssDNA-CNT. The insets show the solution concentrations.

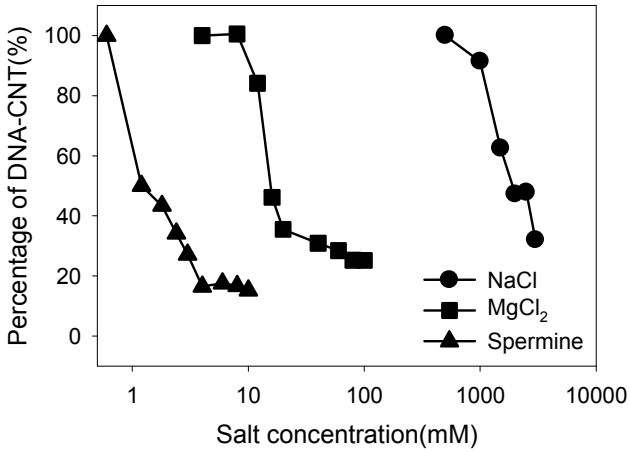


Figure S4. Differential sedimentation assay of salt-induced ssDNA-CNT condensation. Shown are the normalized percentages of ssDNA-CNT in the supernatant as functions of added salt concentration. Noting that the x-axis is logarithmic. The inset shows the salt type. The salt was directly added to the DNA-CNT stock (diluted to 0.5 mg/ml), and the concentration of ssDNA-CNT in the supernatant was measured after spinning at 8000×g for 15 minutes by CNT optical absorption at 990 nm as described in the Materials and Methods section in the main text.

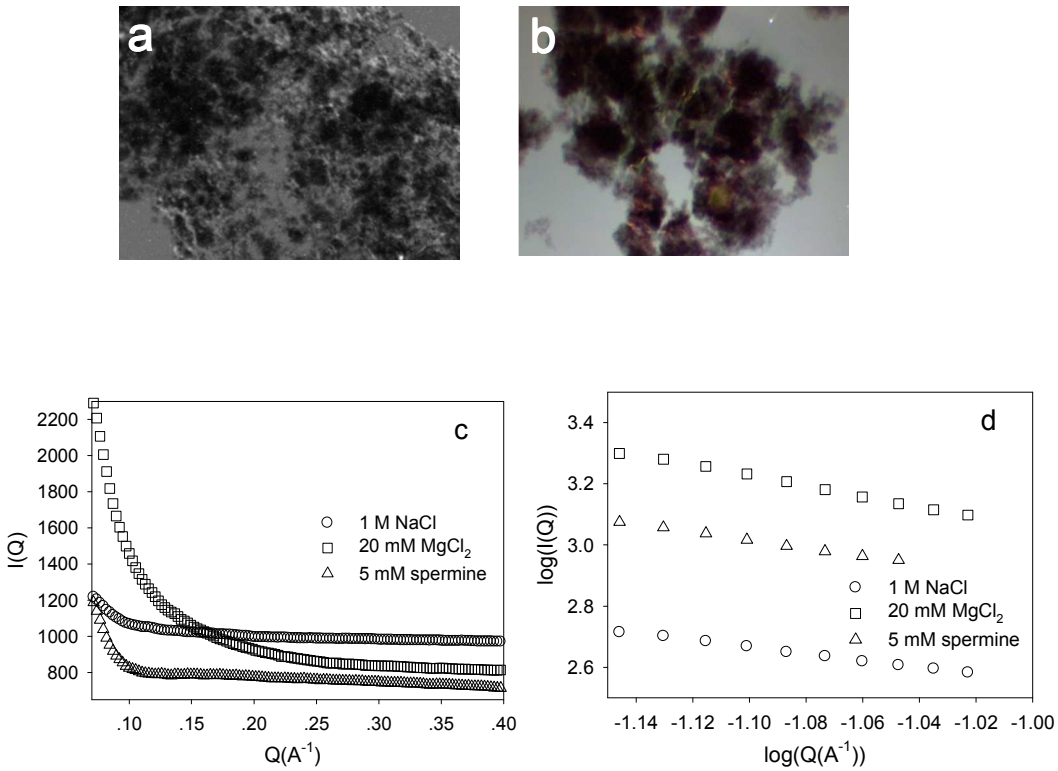


Figure S5. Selective PLM and XRD data of the ssDNA-CNT condensates at varied salts. (a) PLM images of ssDNA-CNT condensed by NaCl; (b) PLM images of ssDNA-CNT condensed by spermine; (c) XRD profiles of the ssDNA-CNT condensates at varied salts. No significant birefringence or XRD peaks are observed. (d) Log-log plot of XRD curves in (c) to calculate fractal dimensions of the formed aggregates. It should be noted that background scattering was not subtracted and such analysis should be taken as being approximate. Good linearity of log(I(Q)) vs log(Q) at low Q region was observed for all curves, and the fractal dimensions (equal to the slopes) were calculated as 1.04, 1.62 and 1.24 in NaCl, $MgCl_2$ and spermine, respectively. This indicates that multivalent ions promote lateral growth of aggregates.

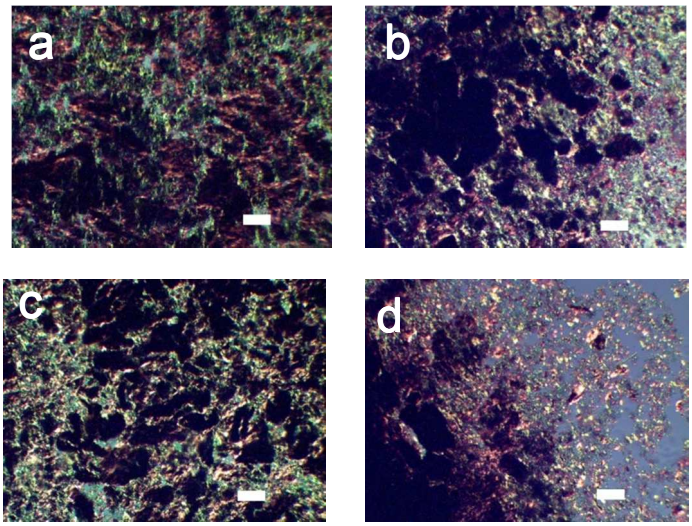


Figure S6. Selective PLM images of DNA-CNT in PEG solutions at varied NaCl. dsDNA-CNT condensates, (a)10% PEG, 150 mM NaCl; (b) 35% PEG, 300 mM NaCl; ssDNA-CNT condensates, (c)10% PEG, 300 mM NaCl; (d) 35% PEG, 300 mM NaCl. The scale bars are 100 μm.

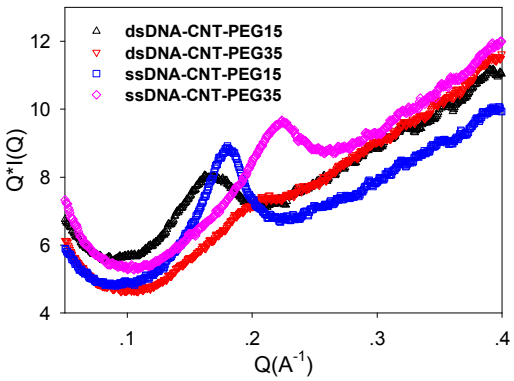


Figure S7. Q*I(Q)~Q plots of DNA-CNT in PEG solutions at 300 mM NaCl. The inset shows the sample information, e.g. “dsDNA-CNT-PEG15” means dsDNA-CNT samples equilibrium in 15% PEG solution.

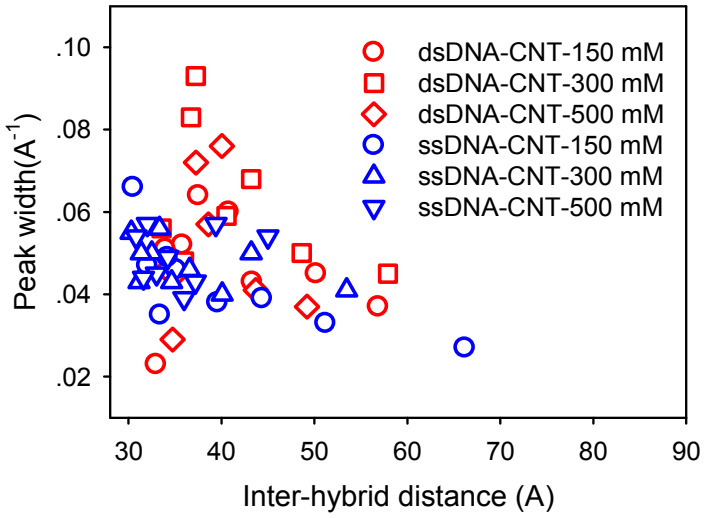


Figure S8. Peak width of XRD peaks from DNA-CNT samples under PEG solutions at varied salt. The inset shows the sample information, e.g. “dsDNA-DNA-150 mM" means “dsDNA-CNT samples at 150 mM NaCl”

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